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Q1. Can I use the Nokia 1200 charger (rated at 350 mAh) for the Nokia 6070 battery which is rated at 850 mAh?

Sharan Dilip
Through e-mail

A1. For charging any battery, you need a charger that supplies higher voltage than the battery which is to be charged. Also, the supply source should be capable of giving the required charging current at that voltage.

Ampere hours (Ah) of a battery show the capacity of the battery. It means that a battery can supply electric energy at its specified voltage for so many hours at the rate of so many amperes. For example, 850mAh rating of a battery means that the battery can give 850mA current at 3.7 volts for one hour, or 425mA current at 3.7 volts for two hours, or 170mA current at 3.7 volts for five hours.

So long as the Nokia 1200 charger can supply the required current to charge the Nokia 6070 battery at a voltage higher than 3.7 volts, it will continue to charge it.

Nokia 1200 batteries are usually rated at 700 mAh and Nokia 6070 at 760 mAh. The battery voltage of all mobile phones is around 3.6 volts. What varies is the connector type between different makes of phones. Also, the Ah ratings and charging rates may be different for various chemical compositions for the same voltage and capacity of the battery.

Q2. Why does a flickering CFL start glowing normally when you touch it or go near it?

Saroj K.
Palakkat, Kerala

A2. There are two main parts in a compact fluorescent lamp (CFL): the gas-filled tube (also called 'bulb' or 'burner') and the magnetic or electronic ballast. An electrical current from the ballast flows through the gas, causing it to emit ultraviolet light. The ultraviolet light then excites the phosphor coating on the inside of the tube. This coating emits visible light.

Electronic ballasts contain a small circuit board with rectifiers, filter capacitor and usually two switching transistors connected as a high-frequency resonant series DC-to-AC inverter. The resulting high frequency, around 40 kHz or higher, is applied to the filament of lamp tube, which ionises the gas inside it and initiates lighting of the CFL.

CFLs that flicker when they start, usually have magnetic ballasts. That is why CFLs with electronic ballasts are now much more common.

Integrated lamps combine a tube, an electronic ballast and either an Edison screw or bayonet fitting in a single CFL unit. These lamps allow consumers to replace incandescent lamps easily with CFLs.

CFLs are produced for both AC and DC input. DC CFLs are popular for use in recreational vehicles and off-the-grid housing. Some families in developing countries are using DC CFLs (with car batteries and small solar panels and/or wind generators), to replace kerosene lanterns.

CFL starter circuit. The figure given on page 18 shows the typical starter circuit of a CFL. When you switch on the supply, the bias voltage to the base of transistor T2 reaches through R6, C2 and diac. This causes T2 to conduct. Thereafter, C2 discharges through T2 and the flow of current through this section is blocked by diode D1, as it is not possible to gather enough energy for re-opening of the diac. Now the transistors are excited by the oscillations over small transformer X1. Transformer X1 consists of a ferrite ring with three windings (five to ten coils). The resonant circuit formed by L1, X1, C3 and C5 produces oscillations to power the filaments of the CFL. Thus the tube lights up in resonance with the frequency specified by the capacitance of C3.

Looking at the starter circuit, the probable reason why a flickering CFL starts glowing normally when you touch it or go near it is that it increases the capacitance across C3 and oscillations start even at a lower voltage. Once the tube fires, oscillations are